Q2: Grouped frequency table for Wealthiest People's Ages.

Data (60 values): 49, 57, 38, 73, 81, 74, 59, 76, 65, 69, 54, 56, 69, 68, 78, 65, 85, 49, 69, 61, 48, 81, 68, 37, 43, 78, 82, 43, 64, 67, 52, 56, 81, 77, 79, 85, 40, 85, 59, 80, 60, 71, 57, 61, 69, 61, 83, 90, 87, 74.

Range: Max (90) - Min (37) = 53

Class width: Using 7 classes with a class width of 8.

Class limits	Tally / Count	Frequency (f)
37 – 44	𝍸	5
45 – 52	𝍸 ||||	4
53 – 60	|||| ||||	8
61 – 68	|||| ||||	9
69 – 76	|||| ||||	9
77 – 84	|||| ||||	10
85 – 92	||||	5
Total.		50

Q5: Grouped frequency distribution for NBA Players' weights.

Data: 240, 210, 220, 260, 250, 195, 230, 270, 325, 225, 165, 295, 205, 230, 250, 210, 220, 210, 230, 202, 250, 265, 230, 210, 240, 246, 225, 180, 175, 215, 215, 235, 245, 250, 215, 210, 195, 240, 240, 225, 260, 210, 190, 260, 230, 190, 210, 230, 185, 260

Min: 165 Max: 325 Range = 160

Grouped frequency distribution (class width=20)

Class limits	frequency (f)
165 – 184	3
185 – 204	7
205 – 224	14
225 – 244	12
245 – 264	10
265 – 284	2
285 – 304	1
305 – 324	0
325 – 344	1
Total	50

Q2:

eights.
225,
250,
235,
, 190,

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Q2: Average Money Spent by first-year college students.

Data summary:
- Electronics $ 728
- Clothing $ 141
- Shoes $ 72
- Total spending $ 1,285

Pie chart Angles calculations (percentage x 360°)
Electronics: (728/1285) x 360° = 203.9° (56.6%)

Dorm decor: (344/1285) x 360° = 96.5° (26.8%)

Clothing: (141/1285) x 360° = 39.5° (11.0%)

Shoes: (72/1285) x 360° = 20.2° (5.6%)

Bar graph set up:
- Place the four categories on the horizontal (x) axis and the dollar amounts ($0 to $800) on the vertical (y) axis.

Q5: Source of energy used worldwide.

Data: Petroleum (39.8%)
Coal (23.2%)
Dry natural gas (22.4%)
Hydroelectric (7.0%)
Nuclear (6.4%)
Others (1.2%)

Pareto chart construction:
A pareto chart is a vertical bar graph ordered from highest frequency/percentage to lowest, combined with a cumulative percentage line.

1. Petroleum: 39.8% (Cumulative: 39.8%)
2. Coal: 23.2% (Cumulative: 63.0%)
3. Dry natural gas: 22.4% (Cumulative: (85.4%)
4. Hydroelectric: 7.0% (Cumulative: 92.4%)
5. Nuclear: 6.4% (Cumulative: 98.8%)
6. Others 1.2% (Cumulative: 100%)

Q10:

Q10: Runners' weekly mileage graph data:

Given frequency distribution:

6-10 : 1

11-15 : 2

16-20 : 3

21-25 : 5

26-30 : 4

31÷35 : 3

36-40 : 2

How to draw:

- Histogram - Draw adjacent vertical bars where the width corresponds to class boundaries (eg: 5.5 to 10.5) and heights equal the frequencies.

Frequency polygon - Plot points of the class midpoints with their respective frequencies and connect them with straight lines anchored to the x-axis at both ends.

- Ogive - Compute cumulative frequencies (1, 3, 6, 11, 15, 18, 20) plot them against the upper class boundaries and connect with an increasing smooth / connected curve.

Q13: NFL salaries Distribution.

Given distribution:

39.9 - 42.8 : 2

42.9 - 45.8 : 2

45.9 - 48.8 : 5

48.9 - 51.8 : 5

51.9 - 54.8 : 12

54.9 - 57.8 : 5

Shape Analysis - The distribution is positively skewed (right skewed) because the highest frequency is concentrated in the 51.9 - 54.8 bracket, with frequencies tapering off toward the lower salary ranges while a heavy cluster exists right before the peak.

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Q4: Values from stem-and-leaf diagrams.

left diagram:
stem 5: 1, 3 → 51, 53

stem 10: 2, 4, 4 → 102, 104, 104

stem 15: 0, 0, 1, 4, 4, → 150, 150, 151, 154, 154

stem 20: 3, 3, 4 → 203, 203, 204

stem 25: 1, 1 → 251, 251

Right diagram:
stem 7: 0, 2, 3 → 70, 72, 73

stem 11: 0, 1, 1, 3, 3, 3 → 110, 111, 111, 113, 113, 113

stem 15: 1, 2, 2, 3 → 151, 152, 152, 153

stem 19: 0, 2 → 190, 192.

stem 23: 1 → 231

Q 10: Back to back stem-and-leaf plot. (French vs English)

Data sets:

French: 31, 44, 44, 45, 46, 47, 50, 51, 52, 53, 53, 57, 58, 58, 61, 61, 66, 69, 75, 78.

English: 43, 44, 44, 52, 56, 58, 63, 65, 65, 68, 69, 71, 72, 72, 77, 79, 79, 84, 85, 88.

Back to back table:

French leaves	stem (Tens)	English leaves
1	3	
7, 6, 5, 4, 4	4	4, 4, 3
8, 8, 7, 3, 3, 2, 1, 0	5	8, 6, 2
9, 6, 1, 1	6	9, 8, 5, 5, 3
8, 5	7	9, 9, 7, 2, 2, 1
	8	8, 5, 4

Q 11:

English)

57,

5, 69,

Q 11: Plant varieties growth comparison.

- To construct this back-to-back stem and leaf plot: Extract the stem (tens digit) in the centre column, place variety 1's leaves sorted in ascending order to the left, and variety 2's leaves sorted in ascending order to the right based on the table data provided on page 05.